

Considerăm o populație în care apare o persoană infectată cu un virus. Virusul se răspândește prin contact direct. După un timp, populația se infectează cu acest virus. Din cauze necunoscute, unele persoane rezistă mai bine la virus, dar sunt purtători și răspândesc boala. La alte persoane, virusul suportă modificări și devine mult mai periculos. Ei răspândesc mai departe virusul în formă modificată și dacă nu sunt vaccinați într-un anumit interval de timp, vor muri. Urmărim evoluția acestei epidemii în programul **Brownian**.

Clasele programului: **Brownian.java** (clasa primară), **Individ.java**, **Doctor.java**, **Virus.java**, **DangerException.java**
Resurse: **im.png**

Clasa Brownian.java

```
import java.awt.*;
import java.net.URL;

public class Brownian extends Frame implements Runnable{
    Individ[] indivizi;
    public Doctor[] doctori;
    int vaccin;
    Virus virus;
    Thread th=null;
    public int ww, hh;
    public Image im, im1;
    public Graphics img;
    int dim=10;
    long tm=100;
    public boolean isrun=false;
    Toolkit tool;

    public static void main(String[] args){
        try{
            new Brownian(args[0], args[1], args[2]);
        } catch(ArrayIndexOutOfBoundsException e){
            System.out.println("Nu ati introdus parametrii necesari (nr_indivizi, nr_doctori, unitati_vaccin)!");
        }
    }

    public URL getResource(String path){
        return getClass().getResource(path);
    }

    public Brownian(String Nr_indivizi, String Nr_doctori, String Unitati_vaccin){
        int nr_indivizi = 500;
        int nr_doctori = 10;
        int unitati_vaccin = 100;

        try{nr_indivizi = Integer.parseInt(Nr_indivizi);}
        catch(NumberFormatException e){}
        try{nr_doctori = Integer.parseInt(Nr_doctori);}
        catch(NumberFormatException e){}
        try{unitati_vaccin = Integer.parseInt(Unitati_vaccin);}
        catch(NumberFormatException e){}

        tool = getToolkit();
        Dimension res = tool.getScreenSize();
        ww = res.width;
        hh = res.height;

        im1 = tool.getImage(getResource("im.png"));

        indivizi = new Individ[nr_indivizi];

        int x=0,y=0;
        for(int i=0;i<indivizi.length;i++){
            double b=Math.random();
            int imun=0;
            if(b>.5) imun=(int)(Math.random()*10);
```

```

        indivizi[i]=new Individ(this,imun);
        indivizi[i].x=(int)(Math.random()*ww);
        indivizi[i].y=(int)(Math.random()*hh);
    }

    doctori = new Doctor[nr_doctori];
    for(int i=0;i<doctori.length;i++){
        doctori[i]=new Doctor(this,unitati_vaccin,i);
        doctori[i].x=(int)(Math.random()*ww);
        doctori[i].y=(int)(Math.random()*hh);
    }

    setResizable(false);
    setTitle("Simularea raspandirii unei epidemii");
    setBackground(Color.black);
    setLayout(null);

    MenuBar mb = new MenuBar();
    Menu file = new Menu("File");
    file.add("Start");
    file.add("Stop");
    file.add("-");
    file.add("Exit");
    mb.add(file);
    setMenuBar(mb);

    resize(ww,hh);
    move(0, 0);
    setVisible(true);

    im = createImage(ww,hh);
    img = im.getGraphics();
}

public boolean handleEvent(Event e){
    if(e.id == Event.WINDOW_DESTROY) System.exit(0);
    else if(e.id == Event.ACTION_EVENT && e.target instanceof MenuItem){
        if(e.arg.equals("Start")){
            th = new Thread(this);
            isrun = true;
            th.start();
            return true;
        } else if(e.arg.equals("Stop")){
            isrun = false;
            th.stop();
            th=null;
            img.clearRect(0,0,ww,hh);
            img.setColor(Color.black);
            img.fillRect(0,0,ww,hh);
            repaint();
            return true;
        } else if(e.arg.equals("Exit"))
            System.exit(0);
    }
    return false;
}

public void run(){
    int r=(int)(Math.random()*indivizi.length);
    lanseazaVirus();
    while(isrun){
        img.clearRect(0,0,ww,hh);
        img.setColor(Color.black);
        img.fillRect(0,0,ww,hh);
        for(int i=0;i<indivizi.length;i++){
            if(indivizi[i].isAlive()){
                indivizi[i].setPosition();
                indivizi[i].setAlive();
            }
        }
    }
}

```

```

        raspare();
        for(int i=0;i<indivizi.length;i++){
            if(indivizi[i].isAlive()){
                int x = indivizi[i].x;
                int y = indivizi[i].y;
                if(!indivizi[i].isInfected())
                    img.setColor(Color.yellow);
                else if(indivizi[i].isMalign())
                    img.setColor(new Color(150,0,0));
                else
                    img.setColor(Color.red);

                if(indivizi[i].isInfected() && !indivizi[i].isImun())
                    img.fillRect(x,y,dim,dim);
                else
                    img.fillOval(x,y,dim,dim);
            }
        }
        boolean exista_doctori = false;
        for(int i=0;i<doctori.length;i++)
            if(doctori[i].isActive())
                exista_doctori=true;
        if(exista_doctori)
            moveDoctors();
        isrun=false;
        for(int i=0;i<indivizi.length;i++)
            if(indivizi[i].isAlive())
                isrun=true;

        if(!isrun)
            img.drawImage(im1,(ww-im1.getWidth(this))/2,(hh-im1.getHeight(this))/2,this);
        repaint();
        try{th.sleep(tm);}
        catch(InterruptedException e){}
    }

}

public void paint(Graphics g){update(g);}

public void update(Graphics g){g.drawImage(im,0,0,this);}

void raspare(){
    for(int i=0;i<indivizi.length;i++){
        if(indivizi[i].isAlive()){
            int x=indivizi[i].x;
            int y=indivizi[i].y;
            for(int j=i+1;j<indivizi.length;j++){
                if(indivizi[j].isAlive()){
                    int u=indivizi[j].x;
                    int v=indivizi[j].y;
                    if(Math.abs(x-u)<=dim && Math.abs(y-v)<=dim){
                        if(!indivizi[i].isInfected() && !indivizi[j].isInfected())
                            continue;
                        if(!indivizi[i].isInfected()){
                            indivizi[i].infect(indivizi[j].getVirus());
                            continue;
                        }else{
                            if(indivizi[j].isMalign()){
                                indivizi[i].decreaseImuninte();
                                continue;
                            }
                        }
                    }
                    if(!indivizi[j].isInfected()){
                        indivizi[j].infect(indivizi[i].getVirus());
                        continue;
                    }else{
                        if(indivizi[i].isMalign()){
                            indivizi[j].decreaseImuninte();
                            continue;
                        }
                    }
                }
            }
            Virus virus=indivizi[i].getVirus();
            indivizi[i].infect(indivizi[j].getVirus());
        }
    }
}

```

```

                                indivizi[j].infect(virus);
                                if(indivizi[j].isMalign())
                                    indivizi[i].decreaseImuninte();
                                if(indivizi[i].isMalign())
                                    indivizi[j].decreaseImuninte();
                                }
                            }
                        }
                    }
                }
            }

void moveDoctors(){
    for(int i=0;i<doctori.length;i++){
        if(doctori[i].isActive()){
            doctori[i].setPosition();
            int u=doctori[i].x;
            int v=doctori[i].y;
            img.setColor(Color.white);
            img.fillRect(u,v,2*dim,2*dim);
            img.setColor(Color.red);
            img.drawLine(u+5,v+dim,u+2*dim-5,v+dim);
            img.drawLine(u+5,v+dim+1,u+2*dim-5,v+dim+1);
            img.drawLine(u+dim,v+5,u+dim,v+2*dim-5);
            img.drawLine(u+dim+1,v+5,u+dim+1,v+2*dim-5);
            for(int j=0;j<indivizi.length;j++){
                if(indivizi[j].isAlive()){
                    int x=indivizi[j].x;
                    int y=indivizi[j].y;
                    if(Math.abs(x-u)<=2*dim && Math.abs(y-v)<=2*dim && indivizi[j].isInfected()){
                        doctori[i].vaccineaza();
                        if(!indivizi[j].isVaccinated())
                            indivizi[j].vaccin();
                    }
                }
            }
            img.setColor(Color.white);
            if(doctori[i].avertisement)
                img.drawString("Atentie! Mai sunt "+doctori[i].getVaccin()+" doze de vaccin!",u+2*dim,v+2*dim);
        }
    }
}

void lanseazaVirus(){
    Virus virus=new Virus("Gripa Galactica");
    int index=(int)(Math.random()*indivizi.length);
    indivizi[index].infect(virus);
}

}

```

===== **Clasa Individ.java** =====

```

import java.util.Date;

public class Individ{
    public int x,y;
    private Virus virus;
    private boolean infected=false;
    private boolean vaccinated=false;
    private int imun;
    private boolean alive=true;
    private long t0,prag1,prag2;
    Brownian brow;

    public Individ(Brownian brow,int imun){
        this.brow=brow;
        this.imun=imun;
        prag1=(int)(Math.random()*120000);
    }
}

```

```

        prag2=(int)(Math.random()*240000);
    }

    public void setPosition(){
        x+=-25+(int)(Math.random()*50);
        y+=-25+(int)(Math.random()*50);
        if(x<0)x=0;
        if(y<0)y=0;
        if(x>brow.ww)x=brow.ww;
        if(y>brow.hh)y=brow.hh;
    }

    public void infect(Virus virus){
        this.virus=virus;
        this.infected=true;
        t0=(new Date()).getTime();
    }

    public Virus getVirus(){return virus;}

    public boolean isInfected(){return infected;}

    public boolean isVaccinated(){return vaccinated;}

    public void vaccin(){
        vaccinated=true;
        t0=(new Date()).getTime();
    }

    public int getImunite(){return imun;}

    public boolean isImun(){return imun>0;}

    public boolean isMalign(){return imun==0;}

    public boolean isAlive(){return alive;}

    public void setAlive(){
        if(imun==0){
            long t=(new Date()).getTime()-t0;
            if(t>prag1)alive=false;
        }else if(vaccinated){
            long t=(new Date()).getTime()-t0;
            if(t>prag2)imun+=10;
        }
        if(imun>15){
            infected=false;
            virus=null;
        }
    }

    public void decreaseImunite(){
        if(imun>0)
            imun--;
    }

}

```

=====

Clasa Doctor.java

=====

```

import java.awt.Color;

public class Doctor{
    public int x,y;
    Brownian brow;
    private int vaccin;

```

```

private boolean activ=false;
public boolean avertisment=false;
int t=10;
int index;

public Doctor(Brownian brow, int vaccin, int index){
    this.brow=brow;
    this.vaccin=vaccin;
    this.index=index;
    if(vaccin>0) activ=true;
}

public void setPosition(){
    x+=-25+(int)(Math.random()*50);
    y+=-25+(int)(Math.random()*50);
    if(x<0)x=0;
    if(y<0)y=0;
    if(x>brow.ww)x=brow.ww;
    if(y>brow.hh)y=brow.hh;
}

public void vaccineaza(){
    if(vaccin>0) vaccin--;
    if(vaccin<10){
        try{danger();}
        catch(DangerException de){
            t--;
            if(t>0){
                System.out.println(de.getMessage());
                de.reactie();
            }
        }
    }
}

public void danger() throws DangerException{
    if(vaccin>0)avertisment=true;
    else{
        activ=false;
        throw(new DangerException(brow,index));
    }
}

public int getVaccin(){return vaccin;}

public boolean isActive(){return activ;}

public void setActive(boolean activ){this.activ=activ;}
}

```

```

=====
Clasa Virus.java
=====

```

```

public class Virus{
    private String virus;
    public Virus(String virus){this.virus=virus;}
    void setVirus(String virus){this.virus=virus;}
    String getVirus(){return virus;}
}

```

```

=====
Clasa DangerException.java
=====

```

```

import java.awt.*;

public class DangerException extends Exception{

```

```

private Brownian brow;

public DangerException(Brownian brow,int index){
    super("Doctorul "+index+" nu mai are vaccin!");
    this.brow=brow;
}

public void reactie(){
    brow.img.clearRect(0,0,brow.ww,brow.hh);
    brow.img.drawImage(brow.im1,(brow.ww-brow.im1.getWidth(brow))/2,(brow.hh-brow.im1.getHeight(brow))/2,brow);
    brow.repaint();
}
}

```

```

=====
start.BAT
=====

```

```
java Brownian nr_indivizi nr_doctori unitati_vaccin
```

De exemplu:

```
java Brownian 500 10 100
```